



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	GREEN CHEMISTRY				
Type of Course	ELECTIVE IN MAJOR				
Semester	V				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. Proficiency in chemistry principles and terminology. 2. Familiarity with environmental issues. 3. Understanding of synthetic organic chemistry principles and methods.				
Course Summary	The course provides a comprehensive overview of the fundamental principles, techniques, and applications of Green Chemistry, empowering students with the knowledge and skills to address environmental challenges through the adoption of sustainable chemical practices				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understanding of fundamental principles of green chemistry and their socio-environmental significance.	U	F	Instructor-created exams / Quiz /Assignment
CO2	Ability to employ alternative starting materials and green reagents in chemical processes, emphasizing sustainability.	Ap	C	Class test /Assignment /Quiz
CO3	Understand and apply knowledge about green solvents and catalysts in the context of sustainable chemical practices.	U	P	Assignment/ Class test

CO4	Apprehend the role of Green Energy and techniques such as microwave and ultrasound assisted reactions in environment-friendly chemical reactions.	An	P	Assignments /Seminar presentation
CO5	Evaluate the practical applications and limitations of green chemistry, and its influence on the world.	E	P	Assignments /Seminar presentation
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Introduction to Green Chemistry		12	20
	1	Green Chemistry - Some important environmental laws (The water (prevention and control of pollution) act, 1974, The environmental protection act of 1986, The air (prevention and control of pollution) act 1981) pollution prevention Act of 1990.	2	
	2	Emergence of green chemistry	1	
	3	Need for Green Chemistry (Prevention and minimization of hazardous products, Reduction of chemical waste and byproducts).	2	
	4	Goals of Green Chemistry. Anastas' twelve principles of green chemistry	2	
	5	Detailed explanation of each postulate with suitable examples	5	
II	Alternative starting materials and reagents in green chemistry		12	20
	6	Use of renewable starting materials: Illustrate with examples such as biodiesel, bioethanol	3	
	7	Polymers from renewable resources.	2	
	8	Green synthesis using Dimethyl carbonate	2	
	9	Green oxidants [Hydrogen peroxide (H ₂ O ₂), Oxygen(O ₂)]	3	
	10	Photochemical synthesis of vitamin D, Advantages compared to conventional synthesis	2	

III	Green solvents and catalysts		12	20
	11	Ionic liquid -Definition and design.	2	
	12	Use of ethyl ammonium nitrate, Ethyl-3-methylimidazolium (EMIM) Chloride and EMIM dicyanamide	2	
	13	Green synthesis using water as solvent	1	
	14	Green synthesis using supercritical carbon dioxide as solvents	1	
	15	Solid state synthesis	2	
	16	Comparison of green solvents and conventional organic solvents. Green catalysis	2	
	17	Biocatalysis and photocatalysis.	2	
IV	Green Energy and techniques		12	20
	18	Mechanism of microwave assisted reaction	2	
	19	Microwave assisted solvent free synthesis of copper phthalocyanine	1	
	20	Microwave assisted reactions in water (Hofmann Elimination, methyl benzoate to benzoic acid and Decarboxylation reaction).	4	
	21	Mechanism of ultrasound assisted reactions, sonochemical Simmons-Smith Reaction (Ultrasonic alternative to Iodine)	3	
	22	Comparison of green and conventional method for one important molecule (oxidation of toluene to benzoic acid by microwave assisted method)	2	
V	Open Ended		12	20
	I	Click chemistry, waste management, renewable energy, can suggest experiments, awareness about presidential green chemistry awards, Limitations of green chemistry.	12	

References:

1. Ahluwalia, V.K. and Kidwai, M.R. New Trends in Green Chemistry, Anamalaya Publishers, 2005
2. Anastas, P.T, Warner, J.K. Oxford Green Chemistry -Theory and Practical, University Press, 1998
3. Matlack, A.S. Introduction to Green Chemistry, Marcel Dekker, 2001
4. Cann, M.C. and Connely, M.E. Real-World Cases in Green Chemistry, American Chemical Society, Washington, 2000
5. Ryan, M.A. and Tinnesand, M., Introduction to Green Chemistry, American Chemical Society, Washington, 2002

6. Lancaster, Mike, Green Chemistry an Introductory Text 2nd Ed., RSC Publishing,. ISBN: 9781-84755-873-2
7. Anastas, P.T and Warner, J.C. Green Chemistry: Theory and Practice, Oxford University Press, 1998
8. Kirchoff, M. and Ryan, M.A. Greener approaches to undergraduate chemistry experiment. American Chemical Society, Washington DC, 2002
9. Ryan, M.A. Introduction to Green Chemistry, Tinnesand; (Ed), American Chemical Society, Washington DC, 2002.

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO 4	PS O5	PS O6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	1						3		1		1	3	
CO 2	1						3		1		3	2	2
CO 3						3	3		1		1	3	
CO 4						3	3		1		3	3	2
CO 5			3				3		1		1	3	3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/Pr actical Exam	Assignmen t /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓